

LESSON 4.8

APPLYING EXPONENT LAWS TO ALGEBRAIC EXPRESSIONS – PART 2



LESSON INTRODUCTION

MA.8.AR.1.1 Apply the laws of exponents to generate equivalent algebraic expressions, limited to integer exponents and monomial bases.

LEARNING TARGETS

- I can write an equivalent expression by applying quotient of powers or power of a quotient laws of exponents.

BENCHMARK COHERENCE

BUILDING ON

MA.7.NSO.1.1 Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational bases.

WORKING TOWARD

MA.912.NSO.1.2 Generate equivalent algebraic expressions using the properties of exponents.

ESSENTIAL QUESTIONS

- How do you apply the quotient of powers law to write equivalent algebraic expressions?
- How do you apply the power of a quotient law to write equivalent algebraic expressions?

LESSON NARRATIVE

In this lesson, students engage in completing, matching, and generating equivalent algebraic expressions by applying the quotient of powers or power of quotient law of exponents.

COMPONENT SUMMARY

4.8.1 WARM-UP (~5 MIN)

Formulate a question that can be solved about the world's largest hamburger

4.8.3 TRY IT (10-15 MIN)

Match equivalent algebraic expressions involving division

4.8.5 WRAP-UP (~5 MIN)

Analyze student work to identify and correct a student's error

4.8.2 GUIDED INSTRUCTION (15 MIN)

Write equivalent algebraic expressions involving division using exponent laws

4.8.4 YOUR TURN! (10-15 MIN)

Use the quotient of powers or power of a quotient law to write equivalent algebraic expressions

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Quotient of powers
- Power of a quotient

MATERIALS

- (Optional) Highlighters or colored pencils

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may confuse which terms are divided and which are subtracted when applying the laws of exponents to expressions with numerical and variable bases.

THINGS TO CONSIDER

- If students forget the laws of exponents, prompt them to write the problems in expanded form and separate the quotient into products of quotients with the same, one base. For example: $\frac{4x^2y^3}{12xy} = \frac{4}{12} \cdot \frac{xx}{x} \cdot \frac{yyy}{y}$.

LITERACY AND LANGUAGE ROUTINES

Notice & Wonder

Engage students in this routine during 4.8.1 Warm-Up. Prompt students to first record or share what they notice about the world's largest hamburger. Then, ask them what they wonder. Finally, prompt students to record or share (if time permits) math problems related to the context.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.5.1 Patterns and Structure

4.8.2 Guided Instruction engages students to discuss the patterns and structure of equivalent algebraic expressions that involve division represented in different ways. Students discover the quotient of powers law of exponents can be applied to generate equivalent expressions involving division with algebraic expressions.

DIFFERENTIATE INSTRUCTION

4.8.2 Guided Instruction provides a visual entry point by having students highlight, circle, or underline each step they take to write an equivalent algebraic expression.

Provide an auditory entry point by asking students to read the completed statements from question 5b aloud to their partner.

While doing 4.8.3 Try It, consider allowing students to use tools to mark the terms they combined to identify the equivalent expression. Consider having students verbally explain to a partner how they matched a couple of the expressions. As part of their explanation, students should include which exponent law(s) they used.

Additional differentiation opportunities are denoted throughout the lesson guide.

$$\frac{16x^9y^6}{8xy^3}$$

4.8.1 WARM-UP: NOTICE AND WONDER

TEACHER MOVES

Notice and Wonder

Before students answer question 1, consider utilizing a Notice and Wonder routine to offer students the opportunity to talk through and hear what their peers notice about the picture. This will focus and solidify their thinking to be able to write a math problem for question 1b.



4.8.1 Warm-Up: Notice and Wonder

1. The world's largest burger belongs to Mallie's Sports Grill and Bar in Detroit and it costs \$8,000.



- a. What do you wonder about the world's largest burger?
Answers will vary. I wonder how much it weighs, what its diameter is, what its circumference is, and how many people it feeds.
- b. Write a math problem that could be solved using the world's largest hamburger.
Answers will vary. If the hamburger feeds 200 people, what is the cost per person?



4.8.2 Guided Instruction: Equivalent Algebraic Expressions Involving Division

- For each expression, rewrite the expression in expanded form. Then, write an equivalent algebraic expression using the fewest possible variable bases.

Given Expression	Expanded Form	Equivalent Expression
$\frac{z^8}{z^6}$	$\frac{z \cdot z \cdot z \cdot z \cdot z \cdot z \cdot z \cdot z}{z \cdot z \cdot z \cdot z \cdot z \cdot z}$	z^2
$\frac{x^4y^7}{xy^5}$	$\frac{x \cdot x \cdot x \cdot x \cdot y \cdot y \cdot y \cdot y \cdot y \cdot y \cdot y}{x \cdot y \cdot y \cdot y \cdot y \cdot y}$	x^3y^2

- Show how each of the equivalent expressions could have been generated by applying the quotient of powers law of exponents.

Given Expression	Apply Quotient of Powers Law of Exponents	Equivalent Expression
$\frac{z^8}{z^6}$	z^{8-6}	z^2
$\frac{x^4y^7}{xy^5}$	$x^{4-1}y^{7-5}$	x^3y^2

The quotient of powers law of exponents can be applied to generate equivalent expressions involving division with algebraic expressions.

When dividing algebraic expressions, the coefficients are divided as well. In both expressions above, the coefficients of the divisor and the dividend are both 1, so the coefficient of the resulting expression is also 1 because $1 \div 1 = 1$.

- Consider the expression $\frac{16x^9y^6}{8xy^3}$. Generate an equivalent expression using division.

$$\frac{16x^9y^6}{8xy^3} = 2x^8y^3$$

- The coefficients will not always be integers. Consider the value of $3 \div 12$.
 - Express the value of $3 \div 12$ as both a fraction and a decimal.

Fraction	$\frac{1}{4}$
Decimal	0.25

- Generate an equivalent expression to $-3m^6p^5 \div 12m^4p^2$.

$$-\frac{1}{4}m^2p^3$$

- Consider the expressions in the table.
 - For each expression, rewrite the expression in expanded form. Then, write an equivalent algebraic expression using the fewest possible variable bases.

Given Expression	Expanded Form	Equivalent Expression
$\left(\frac{w}{n}\right)^2$	$\left(\frac{w}{n}\right)\left(\frac{w}{n}\right)$	$\frac{w^2}{n^2}$
$\left(\frac{q}{-r}\right)^5$	$\left(\frac{q}{-r}\right)\left(\frac{q}{-r}\right)\left(\frac{q}{-r}\right)\left(\frac{q}{-r}\right)\left(\frac{q}{-r}\right)$	$\frac{q^5}{(-r)^5}$
$\left(\frac{-2p}{x}\right)^3$	$\left(\frac{-2p}{x}\right)\left(\frac{-2p}{x}\right)\left(\frac{-2p}{x}\right)$	$\frac{(-2)^3p^3}{x^3}$

4.8.2 GUIDED INSTRUCTION: EQUIVALENT ALGEBRAIC EXPRESSIONS INVOLVING DIVISION

TEACHER MOVES

Based on how students did with the prior lesson, consider having students work on this first question in partners.

DIFFERENTIATE INSTRUCTION

Encourage students to use colored pencils or highlighters to mark terms in the numerator and denominator that divide to one. This can be done by using slash marks or by circling terms that are in the numerator and denominator that divide to one and cancel.

TEACHER MOVES

Students may struggle to recognize (or remember) that when a variable does not have an exponent notated, the exponent is one. For example, $\frac{x^4}{x}$ is $\frac{x^4}{x^1}$ resulting in $x^{4-1} = x^3$.

TEACHER MOVES

Teacher should model that the expression can be written $-\frac{1}{4}m^2p^3$ OR $-0.25m^2p^3$.

Students should not automatically be told to write as decimals because they will not be able to if there is a repeating decimal.

4.8.2 GUIDED INSTRUCTION: EQUIVALENT ALGEBRAIC EXPRESSIONS INVOLVING DIVISION (CONTINUED)

DIFFERENTIATE INSTRUCTION

Provide an auditory entry point by asking students to read the completed statements for question 5b aloud to their partner.

b. Complete the statements.

When a quotient is raised to a power, the

- ☐ power of a quotient
- ☐ power of a product
- ☐ quotient of a power

law of exponents can be

applied by raising both the dividend and divisor to the

same power. If either the dividend or divisor includes a

product, the

- ☐ power of a quotient
- ☐ power of a product
- ☐ quotient of a power

law of exponents

can also be applied. For example, $\left(\frac{-2p}{x}\right)^3$ can be

expressed $\frac{(-2)^3 p^3}{x^3}$, which is equivalent to $\frac{-8 p^3}{x^3}$.



4.8.3 Try It: Equivalent Algebraic Expressions Involving Division

1. Match the expressions in the column on the left to an equivalent expression in the column on the right by drawing arrows.

A. $\frac{-3a^4b^6}{-27a^2b}$	I. $-9a^5b^2$
B. $\frac{27a^4b^6}{3ab^2}$	II. $-\frac{1}{9}a^4b^3$
C. $\frac{27a^6b^4}{-3ab^2}$	III. $9a^3b^4$
D. $\frac{-27a^6b^4}{3a^6b^4}$	IV. $\frac{1}{9}a^3$
E. $\frac{27a^2b^4}{-3a^2b}$	V. -9
F. $\frac{3a^6b^4}{-27a^2b}$	VI. $\frac{1}{9}a^2b^5$
G. $\frac{-3a^7b^8}{-27a^4b^8}$	VII. $-9b^3$

4.8.3 TRY IT: EQUIVALENT ALGEBRAIC EXPRESSIONS INVOLVING DIVISION

DIFFERENTIATE INSTRUCTION

To provide a kinesthetic entry point, the expressions could be written or printed on card stock, and students could work collaboratively to match the expressions.

ENRICHMENT

- How did you determine the coefficient?
- How did you determine the power of each variable?
- Which exponent laws did you use?

4.8.4 YOUR TURN!

DIFFERENTIATE INSTRUCTION

For additional scaffolding, have students write the expressions in expanded form and separate the quotient into products of quotients with the same, one base.

TEACHER MOVES

Consider prompting students to check their answers with a partner after they first attempt the problems independently. Monitor while students are working to determine if there are any questions that need to be debriefed before moving to the next component. This could include highlighting student voices to share rich discussions or comments overheard while they collaborated for the whole class to hear or to address common misconceptions that were evident among the majority of the class.



4.8.4 Your Turn!

- For each expression, apply exponent laws to generate an equivalent expression with the fewest factors possible.

a. $\frac{m^9}{m^4} = m^5$

b. $\frac{5x^4y^7}{15xy^8} = \frac{1}{3}x^3y^2$

c. $\frac{-21c^5d^6}{7c^4d^2} = -3cd^4$

d. $\left(\frac{z}{2}\right)^3 = \frac{z^3}{8}$

e. $\left(\frac{f}{g}\right)^8 = \frac{f^8}{g^8}$

f. $\frac{a^7b^9}{a^6b} = ab^8$



4.8.5 Wrap-Up: Error Analysis

1. Luis rewrote the expression $\frac{-8q^6r^8}{2q^3r^2}$, but he made an error and ended up with an expression that is not equivalent. His work is shown.

$$\frac{-8q^6r^8}{2q^3r^2} = -4q^2r^4$$

- a. What error did Luis make?
Luis divided the exponents instead of subtracting them.
- b. Write a brief note to Luis explaining how to avoid making the same error in the future. Be sure to include what he did correctly in your note.
Answers will vary. You correctly divided the coefficients but made a mistake with the powers of variables. You should have subtracted the exponents of the variables instead of dividing them. For example, $q^{6-3} = q^3$ and $r^{8-2} = r^6$.

4.8.5 WRAP-UP: ERROR ANALYSIS

DIFFERENTIATE INSTRUCTION

Consider giving students the option to respond to question 1 in written form or verbally to a partner. Having students explain correcting the error versus just fixing the error allows them to verbalize the process they would use.